

QUESTION 1 (10 Marks)

a) Evaluate $\frac{\left(\frac{1}{6}\right)^3 + \left(\frac{4}{9}\right)^2}{\left(\frac{2}{3}\right)^4}$ (Give your answer correct to two decimal places)

b) Solve $x^2 + 6x = 91$

c) Solve $|4 - 2x| = 6$

d) If $S = ut + \frac{1}{2}at^2$, make the subject 'a'

e) Find a and b if $(\sqrt{2} + 3\sqrt{3})^2 = a + b\sqrt{6}$

QUESTION 2 (10 Marks)

The line $x - y + 4 = 0$ passes through the points A (-1, 3) and B (3, 7). Find:

- the distance from A to B in exact form
- the distance of the point C (0, 0) to the line $x - y + 4 = 0$ in exact form
- the area of the triangle ABC
- the co-ordinates of the midpoint M between A and B
- the equation of the line through M perpendicular to $x - y + 4 = 0$

QUESTION 3 (10 Marks)

- a) Differentiate
- $(1 - x^2)^5$
 - $\log_e x^2$
 - $\cos 4x$

- b) For the curve with equation $y = \log_e (x - 1)$, state the largest possible domain

- c) Evaluate
- $\int_0^\pi (2\sin x - \sin 2x) dx$
 - $\int_1^4 \frac{1}{2x + 1} dx$
(Give your answer correct to 3 decimal places)
 - $\int_0^1 (e^x - 1)^2 dx$
(Give your answer correct to 3 decimal places)

QUESTION 4 (10 Marks)

- a) Which term of the geometric sequence $\frac{1}{9}, \frac{1}{3}, 1, \dots$ is 2187?
- b) i) Write down a single expression for the sum of the first n terms of the series
 $1 + x^2 + x^4 + x^6 + \dots$
- ii) What is the sum to infinity when $x = \frac{1}{3}$?

c) Find the first term and the common difference of the arithmetic series in which the 29th term is 40 and the sum of the first 8 terms is 26.

QUESTION 5 (10 Marks)

- a) Solve for x
- i) $\log_5\left(\frac{1}{125}\right) = x$
- ii) $\log_{10} x = \log_{10} 8 + \log_{10} 3 - \log_{10} 2$
- b) An urn contains six balls, three red, two blue and one yellow. If three balls are chosen from the urn without replacement, what is the probability that:
- i) the first ball drawn is red?
- ii) all three are red?
- iii) not one is red?
- c) A biased coin has a probability of $\frac{2}{3}$ of showing a head when tossed. Use a tree diagram or otherwise to find the probability of the coin showing at least two heads in three tosses

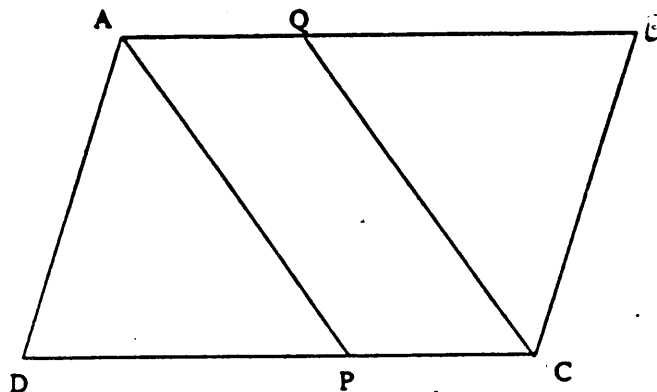
- a) ABCD is a parallelogram.

AP bisects $\angle DAB$ and

CQ bisects $\angle BCD$.

Prove that

- i) $\triangle DAP$ is congruent to $\triangle BCQ$
- ii) $AQ = CP$



- b) In the figure, OA and OB are radii of length 10 cm,

of a circle with centre O. The arc AB of the circle subtends an angle of $\frac{\pi}{3}$ radians at O.

AB is a chord of the circle.

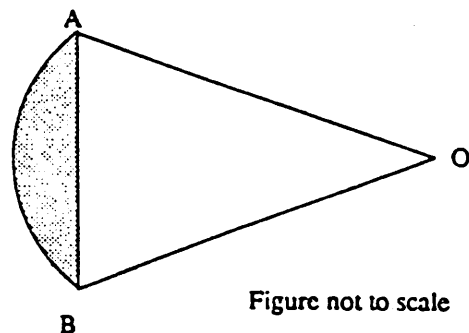


Figure not to scale

- i) Calculate the area of sector AOB
- ii) Calculate the area of the triangle AOB
- iii) Find the area of the segment of the circle shaded in the diagram
(Give all answers correct to 3 decimal places)

QUESTION 7 (10 Marks)

- a) If α and β are the roots of $x^2 - 6x + 2 = 0$, find:

- i) the sum and the product of the roots
- ii) the quadratic equation with roots $2\alpha + 1$, $2\beta + 1$

- b) Prove that $x^2 - 3x + 5 > 0$ for all values of x

- c) For the parabola $x^2 - 4x - 8y - 36 = 0$, find:

- i) the co-ordinates of the vertex;
- ii) the focal length;
- iii) the co-ordinates of the focus;
- iv) the equation of the directrix.

- a) The Hyat Superannuation Scheme offers compound interest calculated yearly on money invested according to the formula $A = P \left(1 + \frac{r}{100} \right)^t$, where \$A\$ is the amount \$P\$ has grown to after being invested for \$t\$ years at \$r\$ % p.a. If Hyat advertises that \$10 000 will yield \$170 000 after 25 years, find the value of \$r\$ to the nearest whole number
- b) The Regent Rollover Fund is operated so that the rate of increase of money invested is governed by the equation $\frac{dP}{dt} = kP$, where \$k\$ is a constant. If \$10 000 grows to \$60 500 in 15 years in this fund, find the value of \$k\$ (to 2 decimal places).
- c) Find the time it takes for \$10 000 to grow to \$170 000 in the Regent Fund and compare the Hyat Scheme with the Regent Fund

QUESTION 9 (10 Marks)

- a) A particle moves in a horizontal straight line such that its distance \$x\$ metres from a fixed point \$O\$ at time \$t\$ seconds is given by the equation $x = 5 + 4t - t^2$. Find:
- the initial displacement, velocity and acceleration of the particle
 - when and where the particle is at rest.
 - when the particle is at the origin
 - the average velocity during the 3rd second
- b) The area between the curve $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, the \$x\$-axis and $x = 1$ and $x = 2$ is rotated about the \$x\$-axis. Find the volume of the solid of revolution formed, leaving your answer in exact form.

- a)
- i) Sketch the curve $y = 3 \sin 2x$ (not on graph paper), for $0 \leq x \leq 2\pi$
 - ii) thence solve the equation $\sin 2x = \frac{1}{3}$, for $0 \leq x \leq 2\pi$, given that the smallest solution in this domain is 0.17
(give all answers correct to two decimal places)
- b)
- i) Find the stationary points for the curve $y = x e^x$ and determine their nature.
 - ii) Sketch the curve $y = x e^x$, showing stationary points, intercepts and asymptotes
 - iii) Hence or otherwise find the values of k for which the equation $x e^x = k$ has
 - A) 2 solutions
 - B) 1 solution

END OF PAPER

Q1, (a) 1.02

$$(b) x^2 + 6x = 91$$

$$x^2 + 6x - 91 = 0$$

$$(x+13)(x-7) = 0$$

$$x = -13, 7.$$

$$(c) 4 - 2x = 6 \text{ or } -4 + 2x = 6$$

$$-2x = 2 \text{ or } 2x = 10$$

$$x = -1 \text{ or } x = 5.$$

$$(d) S = ut + \frac{1}{2}at^2$$

$$\frac{1}{2}at^2 = S - ut$$

$$at^2 = 23 - 2ut$$

$$a = \frac{23 - 2ut}{t^2}$$

$$(e) a + b\sqrt{6} = (52 + 35\sqrt{3})^2$$

$$= 2 + 27 + 6\sqrt{6}$$

$$= 29 + 6\sqrt{6}.$$

$$\therefore a = 29, b = 6.$$

Q2. (a) length_{AB} = $\sqrt{(7-3)^2 + (3-(-1))^2}$

$$= \sqrt{16+16}$$

$$= 4\sqrt{2}$$

$$(b) \text{ perp. dist} = \frac{|1 \times 0 + (-1) \cdot 0 + 4|}{\sqrt{1^2 + 1^2}}$$

$$= \frac{4}{\sqrt{2}}$$

$$= 2\sqrt{2}.$$

$$(c) \therefore \text{Area} = \frac{1}{2} \cdot 25\sqrt{2} \cdot 4\sqrt{2}$$

$$= 8 \text{ sq. units.}$$

$$(d) \text{ coords } m = \left(\frac{3-1}{2}, \frac{7+3}{2} \right)$$

$$= (1, 5)$$

$$(e) \text{ grad: } x - y + 4 = 0$$

$$y = x + 4$$

$$m = 1.$$

$$\text{perp. grad.} = -1.$$

$$\therefore \text{eqn. through } m:$$

$$y - 5 = -1(x - 1)$$

$$y - 5 = -x + 1$$

$$y + x - 6 = 0.$$

Q3 (a)(i) $\frac{d}{dx}(1-x^2)^5 = 5(1-x^2)^4 \cdot -2x$

$$= -10x(1-x^2)^4.$$

$$(ii) \frac{d}{dx}(\log_e x^2) = \frac{2x}{x^2}$$

$$= \frac{2}{x}.$$

$$(iii) \frac{d}{dx}(\cos 4x) = -4 \sin 4x.$$

$$(b) y = \log_e(x-1)$$

$$\text{base } e: x > 1.$$

$$(c)(i) \int_0^\pi (2 \sin x - \sin 2x) \cdot dx.$$

$$= \left[-2 \cos x + \frac{1}{2} \cos 2x \right]_0^\pi$$

$$= +2 + 0.5 + 2 - 0.5$$

$$= 4.$$

Q3. (ii) $\int_1^4 \frac{1}{2x+1} \cdot dx$

$$= \left[\frac{1}{2} \log(2x+1) \right]_1^4$$
$$= \left(\frac{1}{2} \log 9 - \frac{1}{2} \log 3 \right)$$
$$= \frac{1}{2} \log 3$$
$$= 0.549$$

$$\begin{aligned} \text{(iii)} \quad & \int_0^1 (e^x - 1)^2 \cdot dx \\ &= \int_0^1 (e^{2x} + 1 - 2e^x) \cdot dx \\ &= \left[\frac{1}{2} e^{2x} + x - 2e^x \right]_0^1 \\ &= \left[\frac{1}{2} e^2 + 1 - 2e - \frac{1}{2} e^0 + 0 + 2e^0 \right] \\ &= -1.76... + 2\frac{1}{2} \\ &= 0.758 \end{aligned}$$

Q4. (a) $a = \frac{1}{9}$ $r = 3$.

$$T_n = ar^{n-1}$$
$$2187 = 3^{-2} \cdot 3^n$$
$$2187 = 3^{n-3}$$
$$3^7 = 3^{n-3}$$
$$\therefore n = 10.$$

(b) (i) C.P. $a=1$, $r=x^2$.
 $\therefore S_{\text{C.P. terms}} = 1 \left[\frac{(x^2)^n - 1}{x^2 - 1} \right]$
 $S_n = \frac{x^{2n} - 1}{x^2 - 1}$

(iii), $\theta_{\ell,1} = \frac{1}{3}$, $x^2 = \frac{1}{9} = r$.

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1}{1-\frac{1}{9}} \\ &= \frac{9}{8} \quad (1.125) \end{aligned}$$

(c) $T_{29} = a + 28d$

$$S_8 = 4(2a + 7d)$$

$$\begin{aligned} \therefore 40 &= a + 28d & (1) \\ 26 &= 8a + 28d & (2) \\ \therefore -14 &= 7a \\ a &= -2 \end{aligned}$$

from ①:

$$40 = -2 + 28d$$
$$42 = 28d$$
$$d = 1.5.$$

Q5. (a) (i) $\log_5(5^{-3}) = x$
 $\therefore -3 = x$

(ii) $\log_{10} x = \log_{10} \left(\frac{24}{2} \right)$

$$\therefore x = 12.$$

$$(b) (i) P(R) = \frac{3}{6} = \frac{1}{2}$$

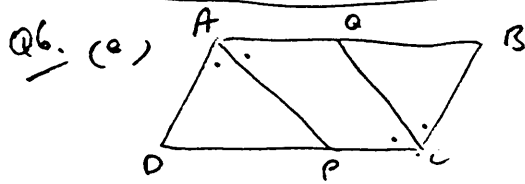
$$(ii) P(R)_{\text{three times}} = \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} = \frac{1}{20}$$

$$(iii) P(\text{not red}) = \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} = \frac{1}{20}$$

(c)

HHH ✓ $\left(\frac{2}{3}\right)^3$
HHT ✓ $\left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)$
HTH ✓
HTT ✓
THH ✓
THT ✓
TTH ✓
TTT ✓

Q5. $\therefore$ total $P = \left(\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2$
 $= \frac{8}{27} + \frac{4}{9}$
 $= \frac{20}{27} \quad (0.740)$



(i) $\angle OAB = \angle BCO$ (opp. $\angle$'s $\parallel$ lines).

$\therefore \angle OAP = \angle BCQ$.

In Δ 's DAP , CBQ :

$\angle OAP = \angle BCQ$ (proved).

$AO = BO$ (opp. sides $\parallel$ lines).

$\angle OPA = \angle PAQ = \angle PCQ = \angle QCB$
 $= \angle BQC$, (alt. $\angle$'s, $DP \parallel AB$ and data)

$\therefore \angle OPA = \angle BQC$.

$\therefore \Delta DAP \equiv \Delta BCQ$ (AAS).

(ii) $\therefore DP = BQ$ (corr. sides
congruent triangles)

$\therefore AQ = PC$ ($AB = DC$, opp.
sides $\parallel$ lines).

(b) (i) Area $\Delta AOB = \frac{1}{2} r^2 \theta$
 $= 50 \frac{\pi}{3}$

(ii) Area $\Delta AOB = \frac{1}{2} r^2 \sin \theta$
 $= 50 \frac{\sqrt{3}}{2}$

(iii) $\therefore$ Area required $= 50 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$
 $= 9.059 \text{ cm}^2$

Q7. (a) (i) $\alpha + \beta = \frac{6}{1}$
 $= 6$
 $\alpha \beta = \frac{2}{1}$
 $= 2$.

(ii) $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

$x^2 - (2 + 2 + 2)x + (2 + 1) \times 2 = 0$

$x^2 - (2 + 2 + 2)x + (4 + 1 + 2 + 2) = 0$

$x^2 - (12 + 2)x + (4 + 1 + 2(6)) = 0$
 $x^2 - 14x + 21 = 0$.

(b) $x^2 - 3x + 5 > 0$ if
 $a > 0$ and $\Delta < 0$.

$a = 1$ $\Delta = b^2 - 4ac$

$b = -3$ $= 9 - 20$

$c = 5$ $= -11$.

$\therefore x^2 - 3x + 5$ is positive
 definite, > 0 for all values of x .

(c) (i) $(x^2 - 4x + 4) = 8y + 36 + 1$
 $(x - 2)^2 = 8(y + 5)$

$\therefore$ vertex $(2, -5)$.

(ii) $"a" = 2$ focal length.

(iii) concave up

$\therefore$ focus is $(2, -3)$.

(iv) directrix: $y = -7$.

(4)

Q 8. (a) $A = P \left(1 + \frac{r}{100} \right)^t$
 $170\,000 = 10\,000 \left(1 + \frac{r}{100} \right)^{25}$
 $17 = \left(1 + \frac{r}{100} \right)^{25}$

$$r = \left(\sqrt[25]{17} - 1 \right) 100.$$

$= 12 \text{ yrs.}$

(b) $\frac{dP}{dt} = kP$
 $P = P_0 e^{kt}$
 $60\,500 = 10\,000 e^{15k}$
 $6.05 = e^{15k}$
 $\frac{\log 6.05}{15} = k.$

$$k = 0.12.$$

(c) $170\,000 = 10\,000 e^{0.12t}$
 $17 = e^{0.12t}$
 $\frac{\log 17}{0.12} = t$

$$t = 23.6 \text{ yrs.}$$

$\therefore$ The Hyet scheme is far superior in generating interest to the Regent Roller Fund.

Q 9. (a) $x = 5 + 4t - t^2$

(i) at $t=0$, $x = 5 \text{ m}$

$$\dot{x} = 4 - 2t$$

at $t=0$, $\dot{x} = 4 \text{ m s}^{-1}$

$$\ddot{x} = -2$$

$\therefore$ at $t=0$, $\ddot{x} = -2 \text{ m s}^{-2}$

(ii) $\ddot{x} = 0$ when $4 - 2t = 0$
 $t = 2 \text{ s.}$

at $t = 2 \text{ s}$, $x = 5 + 8 - 4$
 $= 9 \text{ m.}$

$\therefore$ at rest 9 m from origin after 2 s.

(iii) at origin when $x = 0$

$$0 = 5 + 4t - t^2$$

$$0 = (5 - t)(1 + t)$$

$$\therefore t = 5, (-1 \text{ s}). \therefore \text{at } 5 \text{ s.}$$

(iv) av. vel. = $\frac{\text{vel. at } t=3 - \text{vel. at } t=2}{3 - 2}$

$$= (4 - 6) - (4 - 4)$$

$$= -2 \text{ m s}^{-1}$$

(b) $y = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$

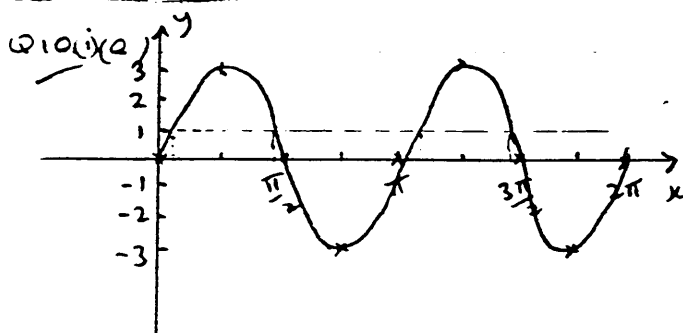
$$\text{Vol} = \pi \int_1^2 y^2 \cdot dx.$$

$$= \pi \int_1^2 \left[x + \frac{1}{x} + 2 \right] \cdot dx.$$

$$= \pi \left[x^{\frac{3}{2}} + \log x + 2x \right]_1^2$$

$$= \pi \left[\frac{4}{\sqrt{2}} + \log 2 + 4 - \frac{1}{\sqrt{2}} - 0 - 2 \right]$$

$$= \pi \left[3^{\frac{1}{2}} + \log 2 \right] \text{ units}^3.$$



Period = π , Amplitude = 3.

(ii) if $\sin 2x = \frac{1}{3}$
 $3 \sin 2x = 1.$

$\therefore$ first simultaneous
 solution of $y = 3 \sin 2x$
 and $y = 1.$

$\therefore$ solution of $0 \leq x \leq 2\pi$
 is $0.17, (\frac{\pi}{2} - 0.17), (\pi + 0.17)$
 and $(\frac{3\pi}{2} - 0.17).$

$= 0.17, 1.40, 3.31, 4.54.$

(b) (i) $y = x e^x$
 stat. pts occur when $\frac{dy}{dx} = 0.$
 $\frac{dy}{dx} = x \cdot e^x + e^x.$
 $= e^x(x+1).$

$\frac{dy}{dx} = 0$ when $e^x = 0$ or
 $x+1=0$

$e^x \neq 0 \therefore$ when $x = -1.$

when $x = -1, y = \frac{-1}{e}$

$\frac{d^2y}{dx^2} = x e^x + e^x + e^x$
 $= e^x(x+2).$

at $x = -1, \frac{d^2y}{dx^2} > 0$

$\therefore (-1, \frac{-1}{e})$ is min.

(ii) at $x = 0, y = 0$

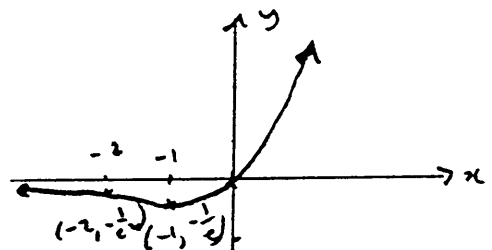
at $y = 0, \therefore x = 0.$

$x \rightarrow +\infty, x \cdot e^x \rightarrow \infty$

$x \rightarrow -\infty, x \cdot e^x$ is
 negative but approaches 0.

$\frac{d^2y}{dx^2} = 0$ when $x+2=0 (e^x \neq 0)$
 $\therefore x = -2$ is
 pt. of inflection $(-2, -\frac{1}{e^2})$

$\frac{d^2y}{dx^2} < 0$ for $x < -2$
 $\frac{d^2y}{dx^2} > 0$ for $x > -2.$



(iii) $\therefore x \cdot e^x = k$ has 2
 (A) solutions for $x < 0$
 and $-\frac{1}{e} \leq k < 0.$

(B) $x \cdot e^x = k$ has 1 solution
 for $x \geq 0$
 and $k \geq 0.$